SIMD

The reason the Cell's SPEs are able to demolish mathematical calculations the way they do is a concept called SIMD—Single Instruction, Multiple Data. The name is suggestive enough, but here's an easy way to understand how this works. If your average, everyday processor was to add a set of ten numbers to another set of ten this is how the instructions flow:

Take the first number from here, take the first number from there, add them.

Take the second number from here, take the second number from there, add them.

... and so on.

Such processors are called *scalar* processors, and are generally suited for everyday applications. However, to perform complex calculations, scientists found this approach painfully slow, which led to the development of the *vector* processor. Instead of operating on only one number at a time, vector processors operate on sets of numbers at a time—one single instruction, but multiple data. So for the Cell's SPE, the above task becomes as simple as:

Take all these numbers, take all those numbers, add them.

This technique lends itself perfectly for processing pixel data for rendering a scene, and more importantly, the physics and AI processing that will be the Cell's responsibility.

Maximum Security Cell

With Sony—known for its fervent efforts to protect its intellectual property—being one of the driving forces behind Cell, it's no surprise that the anti-piracy angle has been well thought up. The Cell will feature application security at the hardware level. Cracking software involves reading what it puts into the system memory, and thus far, it's been the operating system that prevented hackers from doing this. Compromise the OS kernel, however, and all falls to naught. The Cell, however, doesn't let even the OS kernel see what's inside the SPEs' local memory, so the question of hackers peeking into application memory doesn't come up. Voila! Automatic anti-piracy!

Big Picture

Cell isn't just a processor—'tis but a mere piece in the game for World Domination (TM). Cell's software is compiled into little "apulets," which will distribute themselves to all available Cell processors—be it on the same board, the same LAN, even over the Internet. The result is a massive grid computer, where every idle Cell you're connected to becomes a possible candidate to offload computing on. Imagine gaming on your PS3, while your Cell-enabled HDTV and PDA crunch numbers for a research lab in one corner of the Earth trying to find a cure for cancer!

So will we ever see the Cell in our desktops? Quite possibly, but there are a few things about this whole thing that might throw a spanner or two into the machinery. Firstly, there's the PPE. While it's way ahead of the competition for gaming, it will still lose out to current processors when it comes to general-purpose applications. However, throwing in more PPEs on the chip could well turn that around. Secondly, the Cell absolves itself of a lot of the responsibilities of regular processors, instead, offloading them on to the developers who will be writing the compilers for them. The insane task of writing a compiler that can intelligently exploit the Cell and all its features might just put programmers off, and that would be the end of the Cell's PC prospects.

The only people willing to code to that level are game developers, so game consoles, at least, will see a lot more of the Cell. For now, Sony and Toshiba plan to release Cell-based HDTVs and PDAs in the near future, and the Cell will probably conquer the living room before it does the desktop. ■

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